

Day 36 - Project: Enterprise Security Monitoring Architecture Review

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Table of Contents

1. Executive Summary
2. Project Objective
3. Organization Environment Overview
4. Assessment Methodology
5. Asset Visibility Assessment
6. Log Source Assessment
7. MITRE ATT&CK Coverage Assessment
8. Detection Gap Analysis
9. SOC Maturity Assessment
10. Executive Security Review
11. Strategic Security Roadmap
12. ATT&CK Coverage Matrix
13. Detection Gap Matrix
14. SOC Maturity Matrix
15. Executive Presentation Summary
16. Security Improvement Roadmap
17. Recommendations
18. Conclusion

1. Executive Summary

This Enterprise Security Monitoring Architecture Review was conducted to evaluate the effectiveness of the organization's existing monitoring capabilities and identify visibility gaps that could expose critical business systems to cyber threats.

The organization consists of approximately 500 employees operating across Windows endpoints, Linux servers, AWS cloud infrastructure, Microsoft 365 services, VPN infrastructure, and a centralized SIEM platform.

Recent incidents involving phishing attacks, ransomware infections, credential theft attempts, and cloud misuse indicate increasing cyber risk exposure and highlight the need for enhanced monitoring, detection engineering, and incident response capabilities.

The objective of this assessment is to evaluate security visibility across the enterprise, identify monitoring deficiencies, measure ATT&CK coverage, assess SOC maturity, and develop a strategic roadmap for improving security operations.

2. Project Objective

The primary objectives of this assessment are:

- Assess enterprise monitoring effectiveness
- Identify security visibility gaps
- Evaluate detection engineering coverage
- Assess ATT&CK framework coverage
- Review current SOC capabilities
- Improve threat detection and response
- Develop a strategic security roadmap
- Enhance cyber resilience

3. Organization Environment Overview

Workforce

- 500 Employees
- Hybrid work environment
- Remote VPN access enabled

Endpoint Environment

Windows Systems

- User Workstations
- Administrative Systems
- Domain Controllers
- File Servers

Linux Servers

- Application Servers
- Web Servers
- Database Servers
- Backup Infrastructure

Cloud Infrastructure

AWS Environment

Services in use:

- EC2
- S3
- IAM
- CloudTrail

- VPC
- Security Groups

SaaS Infrastructure

Microsoft 365

Services include:

- Exchange Online
- SharePoint Online
- Teams
- OneDrive
- Azure Active Directory

Security Infrastructure

- SIEM Platform
- Firewall
- VPN Gateway
- Antivirus Solutions
- Active Directory

4. Assessment Methodology

The assessment was conducted through seven major phases:

Phase 1 – Asset Visibility Assessment

Evaluate asset inventory completeness and visibility.

Phase 2 – Log Source Assessment

Assess available telemetry and logging capabilities.

Phase 3 – ATT&CK Coverage Assessment

Map current detections against MITRE ATT&CK tactics and techniques.

Phase 4 – Detection Gap Analysis

Identify missing detections and monitoring blind spots.

Phase 5 – SOC Maturity Assessment

Evaluate operational security monitoring maturity.

Phase 6 – Executive Security Review

Assess business-level security risks.

Phase 7 – Strategic Roadmap Development

Develop phased improvement recommendations.

5. Asset Visibility Assessment

Current Asset Inventory

Asset Type	Estimated Count
Windows Endpoints	400
Linux Servers	50
AWS Resources	100
Microsoft 365 Users	500
VPN Users	450

Assessment Findings

Strengths

- Centralized Active Directory
- Managed endpoint infrastructure
- Cloud asset visibility through AWS console
- SIEM deployment in place

Weaknesses

- Limited automated asset discovery
- Incomplete Linux asset inventory
- Limited cloud workload visibility
- Third-party asset tracking deficiencies

Risks

Unmanaged assets may:

- Evade monitoring
- Introduce vulnerabilities
- Enable unauthorized access
- Create security blind spots

Risk Rating

HIGH

6. Log Source Assessment

Current Log Sources

Log Source	Status
Windows Event Logs	Available
Active Directory Logs	Available
VPN Logs	Available
Firewall Logs	Available
Microsoft 365 Audit Logs	Partial

AWS CloudTrail	Partial
Linux Syslogs	Partial
Endpoint Security Logs	Limited
DNS Logs	Missing
Proxy Logs	Missing
DLP Logs	Missing

Assessment Findings

Adequately Monitored

- Authentication Events
- VPN Activity
- Active Directory

Partially Monitored

- Linux Servers
- AWS Environment
- Microsoft 365

Poorly Monitored

- Data Exfiltration
- DNS Activity
- Insider Threat Activity
- Cloud Privilege Abuse

Risk Rating

HIGH

7. MITRE ATT&CK Coverage Assessment

Current Coverage

ATT&CK Tactic	Coverage
Initial Access	Medium
Execution	Medium
Persistence	Low
Privilege Escalation	Medium
Defense Evasion	Low
Credential Access	Medium
Discovery	Low
Lateral Movement	Low
Collection	Low
Exfiltration	Low
Impact	Medium

Key Findings

Strong visibility exists for:

- Authentication monitoring
- Malware detection
- Basic endpoint security

Weak visibility exists for:

- Lateral movement
- Credential theft
- Cloud attacks
- Data exfiltration
- Persistence mechanisms

8. Detection Gap Analysis

Recent Incident Review

Phishing Incident

Detection Status:

Partially Detected

Credential Theft

Detection Status:

Delayed Detection

Ransomware Activity

Detection Status:

Detected During Encryption Stage

Cloud Misuse

Detection Status:

Not Detected

Critical Detection Gaps

PowerShell Abuse

Missing:

- Encoded command detection
- Obfuscated script detection

Credential Dumping

Missing:

- LSASS monitoring
- Mimikatz detection

Lateral Movement

Missing:

- Remote execution detection
- SMB abuse monitoring

Cloud Threats

Missing:

- IAM anomaly detection
- Privilege escalation monitoring

Data Exfiltration

Missing:

- DLP integration
- Large transfer monitoring

Risk Rating: CRITICAL

9. SOC Maturity Assessment

Maturity Model

Capability	Current Level
Log Collection	Level 3
Alerting	Level 3
Detection Engineering	Level 2
Threat Hunting	Level 1
Incident Response	Level 2
Automation	Level 1
Reporting	Level 3

Overall SOC Maturity

Current State:

Level 2 – Developing

Characteristics:

- Reactive monitoring
- Limited threat hunting

- Manual investigations
- Basic detection coverage

Target State:

Level 4 – Managed SOC

Characteristics:

- Advanced detections
- Automated response
- Threat hunting program
- Continuous monitoring

10. Executive Security Review

Executive Risk Summary

Security Area	Risk Level
Endpoint Security	High
Identity Security	Critical
Cloud Security	High
Data Protection	Critical
Monitoring Coverage	High

Business Risks

- Ransomware disruption
- Credential compromise
- Regulatory violations
- Financial loss
- Operational downtime
- Reputation damage

11. Strategic Security Roadmap

Phase 1 (0–3 Months)

Quick Wins

- Enable PowerShell logging
- Deploy Sysmon
- Enable CloudTrail
- Enable Microsoft 365 auditing

Phase 2 (3–6 Months)

Detection Improvements

- Credential theft detection
- Ransomware monitoring
- Cloud activity monitoring
- Privileged account monitoring

Phase 3 (6–12 Months)

SOC Enhancements

- Threat intelligence integration
- UEBA implementation
- Threat hunting program
- SOAR deployment

Phase 4 (12–18 Months)

Advanced Security Operations

- Detection engineering team
- Purple team exercises
- Continuous security validation
- Risk-based alerting

12. ATT&CK Coverage Matrix

ATT&CK Tactic	Current	Target
Initial Access	Medium	High
Execution	Medium	High
Persistence	Low	High
Privilege Escalation	Medium	High
Defense Evasion	Low	High
Credential Access	Medium	High
Discovery	Low	High
Lateral Movement	Low	High
Collection	Low	High
Exfiltration	Low	High
Impact	Medium	High

13. Detection Gap Matrix

Threat	Current Coverage	Gap Severity
Phishing	Medium	Medium
Credential Theft	Low	High

PowerShell Abuse	Low	High
Ransomware	Medium	High
Cloud Misuse	Low	Critical
Insider Threat	Low	Critical
Data Exfiltration	Low	Critical

14. SOC Maturity Matrix

Domain	Current	Target
Monitoring	3	4
Detection	2	4
Response	2	4
Threat Hunting	1	4
Automation	1	4
Reporting	3	4

15. Executive Presentation Summary

Key Findings:

- Multiple monitoring gaps identified
- Critical cloud visibility deficiencies
- Inadequate ransomware detection coverage
- Limited threat hunting capability
- Lack of advanced detection engineering

Business Impact:

- Increased attack surface
- Higher ransomware exposure
- Increased insider threat risk
- Compliance concerns

16. Security Improvement Roadmap

Priority 1:

Identity Protection

Priority 2:

Cloud Security Monitoring

Priority 3:

Advanced Endpoint Detection

Priority 4:

Threat Hunting Capability

Priority 5:

SOC Automation

17. Recommendations

Immediate Actions

- Enable advanced logging
- Deploy EDR/XDR
- Implement MFA
- Improve SIEM correlation

Medium-Term Actions

- Implement UEBA
- Add Threat Intelligence
- Conduct Threat Hunting

Long-Term Actions

- Deploy SOAR
- Establish Detection Engineering Team
- Conduct Purple Team Exercises

18. Conclusion

The Enterprise Security Monitoring Architecture Review identified several critical visibility gaps across endpoint, cloud, identity, and data protection environments. Although the organization has implemented foundational security monitoring through a SIEM platform, current monitoring capabilities remain insufficient to detect advanced cyber threats proactively.

The overall SOC maturity level was assessed as Level 2 (Developing), indicating the need for significant improvements in detection engineering, automation, threat hunting, and cloud security monitoring.

By implementing the recommended strategic roadmap, enhancing ATT&CK coverage, improving log collection, deploying advanced security monitoring technologies, and strengthening SOC operations, the organization can significantly improve cyber resilience, reduce attacker dwell time, and establish a mature enterprise-grade security monitoring program.